Note 1.6 Partition Matrices

Partition Matrices

- **Strategy:**
  To think a matrix as being a composition of submatrices, which can be divided by draw horizontal lines between rows and vertical lines between columns. The small matrices are often called blocks.
- **Partition into columns:**
  In general, if A is an $m \times n$ matrix and B is an $n \times r$ matrix, which is partitioned into columns $(\mathbf{b_1}, \mathbf{b_2}, \cdots, \mathbf{b}_n)$, then the block multiplication of AB is

$$AB = (A\mathbf{b_1}, A\mathbf{b_2}, \cdots, A\mathbf{b}_r)$$

- **Partition into rows:**
  In general, if A is an $m \times n$ matrix and B is an $n \times r$ matrix, in which A is partitioned into rows $\begin{pmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ \cdots \\ \vec{\mathbf{a}}_n \end{pmatrix}$, then the block multiplication of AB is

$$AB = \begin{pmatrix} \vec{\mathbf{a}}_1 B \\ \vec{\mathbf{a}}_2 B \\ \vdots \\ \vec{\mathbf{a}}_m B \end{pmatrix}$$

Block Multiplication

- **Case 1: Partition into two blocks by columns:**
  If $B = (B_1 \quad B_2)$, where B_1 is an $n \times t$ matrix and B_2 is an $n \times (r - t)$ matrix, then

$$A\,(B_1 \quad B_2) = (AB_1 \quad AB_2)$$

  where A is a $m \times n$ matrix.
- **Case 2: Partition into two blocks by rows:**
  If $A = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$, where A_1 is a $t \times n$ matrix and A_2 is a $(m - t) \times n$ matrix, then

$$\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} B = \begin{pmatrix} A_1 B \\ A_2 B \end{pmatrix}$$

- **Case 3: Partition into four blocks:**
  Let $A = (A_1 \quad A_2)$ and $B = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix}$, where A_1 is a $m \times t$ matrix, A_2 is a $m \times (n-t)$ matrix, B_1 is an $t \times r$ matrix and B_2 is an $(n-t) \times r$ matrix, then

$$(A_1 \quad A_2) \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = A_1 B_1 + A_2 B_2$$

- **Case 4: Partition into multiple proper blocks:**
  In general, if the blocks have the proper dimensions, the block multiplication can be carried out in the same manner as ordinary matrix multiplication. Thus, if

$$A = \begin{pmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \\ A_{m1} & \cdots & A_{mn} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} B_{11} & \cdots & B_{1r} \\ \vdots & & \\ B_{n1} & \cdots & B_{nr} \end{pmatrix}$$

  then

$$AB = \begin{pmatrix} C_{11} & \cdots & C_{1r} \\ \vdots & & \\ C_{m1} & \cdots & C_{mr} \end{pmatrix}$$

  where

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

Outer Product Expansions

- **Scalar product(inner product):**
  If a row vector with n rows times a column vector with n columns, or say, if a $1 \times n$ matrix times an $n \times 1$ matrix, the product will be a scalar, then the product is referred as a scalar product or inner product. For example,

$$\mathbf{x}^T \mathbf{y} = (x_1, x_2, \cdots, x_n) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

  where $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$.
- **Outer product:**
  If a column vector with n columns times a row vector with n rows, or say, if an $n \times 1$ matrix times a $1 \times n$ matrix, the product will be an $n \times n$ matrix, then the product is referred as a

outer product. For example,

$$\mathbf{x}^T\mathbf{y} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} (y_1, y_2, \cdots, y_n) = \begin{pmatrix} x_1y_1 & x_1y_2 & \cdots & x_1y_n \\ x_2y_1 & x_2y_2 & \cdots & x_2y_n \\ \vdots & & & \\ x_ny_1 & x_ny_2 & \cdots & x_ny_n \end{pmatrix}$$

- **Outer product expansion:**
  Generalize the idea of outer product to matrix, suppose that there is an $m \times n$ matrix X and an $n \times r$ matrix Y, then

$$XY = (\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n) \begin{pmatrix} \vec{\mathbf{y}}_1 \\ \vec{\mathbf{y}}_2 \\ \vdots \\ \vec{\mathbf{y}}_n \end{pmatrix} = \mathbf{x}_1\vec{\mathbf{y}}_1 + \cdots + \mathbf{x}_n\vec{\mathbf{y}}_n$$

  This representation is referred as an outer product expansion.^99b99a

Section 1.6 Exercises

- Question 9th
  - Since D is a diagonal matrix, then we know $\mathbf{d}_j = d_{jj}\mathbf{e}_j$. Thus

$$D = (d_{11}\mathbf{e}_1, d_{22}\mathbf{e}_2, \cdots, d_{nn}\mathbf{e}_n)$$

  - For AD,

$$\begin{aligned} AD &= A(d_{11}\mathbf{e}_1, \cdots, d_{nn}\mathbf{e}_n) \\ &= (d_{11}A\mathbf{e}_1, \cdots, d_{nn}A\mathbf{e}_n) \\ &= (d_{11}\mathbf{a}_1, \cdots, d_{nn}\mathbf{a}_n) \end{aligned}$$

- Question 11th
  For the matrix A, we can use the $(A|I)$ to find A^{-1}.

$$\begin{aligned} (A|I) &= \left(\begin{array}{cc|cc} A_{11} & A_{12} & I & O \\ O & A_{22} & O & I \end{array}\right) \\ &= \left(\begin{array}{cc|cc} I & A_{11}^{-1}A_{12} & A_{11}^{-1} & O \\ O & I & O & A_{22}^{-1} \end{array}\right) \\ &= \left(\begin{array}{cc|cc} I & O & A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ O & I & O & A_{22}^{-1} \end{array}\right) \end{aligned}$$

  Note that we are through the row operation thus the matrix is left multiplication.
- Question 12th
  Suppose A is singular, then there exists $\mathbf{x}_1 \neq \mathbf{0}$ such that $A\mathbf{x}_1 = \mathbf{0}$. Then let $\mathbf{x} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{0} \end{pmatrix}$, then

$$M\mathbf{x} = \begin{pmatrix} A & O \\ O & B \end{pmatrix} \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{0} \end{pmatrix} = \mathbf{0}$$

since $\mathbf{x} \neq \mathbf{0}$, then M is singular. And B is similar.